WORKED SOLUTION

# STEP I 1989, Question 3

Pure • Vector geometry

> **STEP I 1989, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/89-s1-q3

## The key idea

Put everything in terms of the two vectors $\mathbf{a}$ and $\mathbf{b}$, with $O$ as origin. Then "perpendicular" becomes "scalar product zero", and because $\mathbf{a}\cdot\mathbf{a} = \mathbf{b}\cdot\mathbf{b} = 1$, that scalar product depends only on $x$, $OY$ and the one unknown $\mathbf{a}\cdot\mathbf{b}$. Using $OX + OY = 1$ leaves a single equation linking $x$ and $\mathbf{a}\cdot\mathbf{b}$. The bounds then come from how far $x$ can range.

## Solution

### The condition

Take $O$ as origin and write $k = \mathbf{a}\cdot\mathbf{b}$. Since $C$ is one third of the way from $A$ to $B$, and $D$ two thirds of the way,

$$\overrightarrow{OC} = \mathbf{a} + \tfrac{1}{3}(\mathbf{b}-\mathbf{a}) = \tfrac{1}{3}(2\mathbf{a}+\mathbf{b}), \qquad \overrightarrow{OD} = \tfrac{1}{3}(\mathbf{a}+2\mathbf{b}).$$

Since $X$ is on $OA$ with $OX = x$ and $|\mathbf{a}| = 1$, we have $\overrightarrow{OX} = x\mathbf{a}$. Similarly $\overrightarrow{OY} = y\mathbf{b}$ with $y = OY = 1 - x$. So

$$3\overrightarrow{CY} = (3y-1)\mathbf{b} - 2\mathbf{a}, \qquad 3\overrightarrow{DX} = (3x-1)\mathbf{a} - 2\mathbf{b}.$$

The lines are perpendicular exactly when the scalar product is zero. Expanding, and using $\mathbf{a}\cdot\mathbf{a} = \mathbf{b}\cdot\mathbf{b} = 1$,

$$\begin{aligned} 9\,\overrightarrow{CY}\cdot\overrightarrow{DX} &= (3x-1)(3y-1)k - 2(3y-1) - 2(3x-1) + 4k \\ &= k[(3x-1)(3y-1)+4] - 2(3x+3y-2). \end{aligned}$$

Now use $x + y = 1$. Then $3x + 3y - 2 = 1$, and

$$(3x-1)(3y-1) = 9xy - 3(x+y) + 1 = 9xy - 2.$$

So the perpendicularity condition becomes $k(9xy+2) = 2$. With $y = 1 - x$, this is

$$\boxed{\mathbf{a}\cdot\mathbf{b} = \frac{2}{9x(1-x)+2}}$$

(the denominator is at least 2, so dividing by it is safe).

### The bounds

$X$ lies on the segment $OA$, which has length 1, so $0 \le x \le 1$. (Then $y = 1 - x$ is also in $[0, 1]$, so $Y$ is automatically on $OB$.) On this interval

$$x(1-x) = \tfrac{1}{4} - (x - \tfrac{1}{2})^2,$$

so $0 \le x(1-x) \le \frac{1}{4}$, with the maximum at $x = \frac{1}{2}$ and the minimum at $x = 0$ or 1. Hence

$$2 \le 9x(1-x) + 2 \le \tfrac{9}{4} + 2 = \tfrac{17}{4}.$$

Taking reciprocals reverses the inequalities (everything is positive), and multiplying by 2 gives

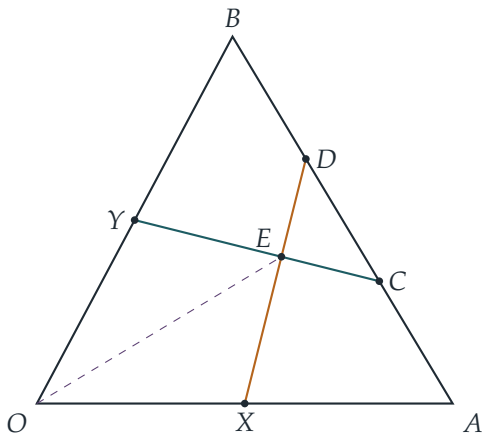
$$\frac{8}{17} \le \mathbf{a} \cdot \mathbf{b} \le 1,$$

as required. (The value 1 would need $\mathbf{a} = \mathbf{b}$, which is not a genuine triangle, so in practice $\mathbf{a} \cdot \mathbf{b} < 1$; the stated inequality still holds.)

### The largest angle and $OE$

**Ask yourself** Why does the largest angle correspond to the *smallest* value of $\mathbf{a} \cdot \mathbf{b}$?

Since $|\mathbf{a}| = |\mathbf{b}| = 1$, $\mathbf{a} \cdot \mathbf{b} = \cos A\hat{O}B$, and cosine is decreasing on $[0, \pi]$. So the angle is largest when $\mathbf{a} \cdot \mathbf{b} = \frac{8}{17}$, that is, when $x = \frac{1}{2}$ (and so $y = \frac{1}{2}$). Then $X$ and $Y$ are the midpoints of $OA$ and $OB$, and the angle is $\arccos \frac{8}{17} \approx 61.9°$.




Now the whole figure is symmetric under swapping $\mathbf{a}$ and $\mathbf{b}$: this swaps $A \leftrightarrow B$, $C \leftrightarrow D$ and $X \leftrightarrow Y$, so it swaps the lines $CY$ and $DX$. Their intersection $E$ is therefore unchanged by the swap, so it lies on the line of symmetry: $\overrightarrow{OE} = t(\mathbf{a} + \mathbf{b})$ for some $t$.

A general point of $DX$ is

$$\overrightarrow{OX} + s\overrightarrow{XD} = \tfrac{1}{2}\mathbf{a} + s(\tfrac{1}{3}\mathbf{a} + \tfrac{2}{3}\mathbf{b} - \tfrac{1}{2}\mathbf{a}) = (\tfrac{1}{2} - \tfrac{s}{6})\mathbf{a} + \tfrac{2s}{3}\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel, we can compare coefficients: this is a multiple of $\mathbf{a}+\mathbf{b}$ when $\frac{1}{2} - \frac{s}{6} = \frac{2s}{3}$, giving $s = \frac{3}{5}$ and $t = \frac{2}{5}$. So $\overrightarrow{OE} = \frac{2}{5}(\mathbf{a}+\mathbf{b})$. Finally

$$|\mathbf{a}+\mathbf{b}|^2 = \mathbf{a}\cdot\mathbf{a} + 2\,\mathbf{a}\cdot\mathbf{b} + \mathbf{b}\cdot\mathbf{b} = 2 + \tfrac{16}{17} = \tfrac{50}{17},$$

so

$$OE = \frac{2}{5}\sqrt{\frac{50}{17}} = \frac{2}{5}\cdot\frac{5\sqrt{2}}{\sqrt{17}} = \frac{2\sqrt{2}}{\sqrt{17}} = \boxed{\frac{2\sqrt{34}}{17}} \approx 0.686.$$

## Checks

- In coordinates, take $\mathbf{a} = (1,0)$ and $\mathbf{b} = (\frac{8}{17}, \frac{15}{17})$. Then $X = (\frac{1}{2}, 0)$, $D = (\frac{11}{17}, \frac{10}{17})$, so $\overrightarrow{XD} = (\frac{5}{34}, \frac{10}{17})$. Also $Y = (\frac{4}{17}, \frac{15}{34})$ and $C = (\frac{14}{17}, \frac{5}{17})$, so $\overrightarrow{CY} = (-\frac{10}{17}, \frac{5}{34})$. Their scalar product is $-\frac{50}{578} + \frac{50}{578} = 0$, as it should be.
- $E = \frac{2}{5}(\frac{25}{17}, \frac{15}{17}) = (\frac{10}{17}, \frac{6}{17})$, and $|OE| = \frac{\sqrt{136}}{17} = \frac{2\sqrt{34}}{17} \approx 0.6860$. This matches a direct numerical intersection of the two lines.
- The value $s = \frac{3}{5}$ lies in $[0,1]$, so $E$ really is between $X$ and $D$, inside the triangle.

## Common slips

- Forgetting that $\mathbf{a}\cdot\mathbf{a} = 1$ (and so leaving $|\mathbf{a}|^2$ terms in the condition), or getting the trisection points the wrong way round.
- Thinking that the largest angle means the largest $\mathbf{a}\cdot\mathbf{b}$: the cosine decreases as the angle grows.
- Writing $\overrightarrow{OX} = x$ rather than $x\mathbf{a}$. The point $X$ is a vector; $x$ is only its length.

## Beyond STEP

The method here, writing every point as a combination of two basis vectors and turning geometry into scalar products, is the start of working with inner products and Gram matrices. A configuration of vectors is fixed up to rotation by its table of scalar products, which here is just 1, 1 and $\mathbf{a}\cdot\mathbf{b}$. That is why the whole question reduces to one number. Strang's *Introduction to Linear Algebra* develops these ideas.

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